



NEO Coordination Centre

A database maintained by ESA based on data collected by JPL on small bodies and their orbital characteristics. <https://dgit.in/nov19-05>



The moment of discovery

An animation of comet C/2019 Q4 captured by the amateur astronomer who discovered it, Gennadiy Borisov. <https://dgit.in/nov19-06>

INTERSTELLAR INTERLOPER 2.0

Barely two years after 'Oumuamua, the Solar System has received its second known visitor from the stars

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On August 30, 2019, a Crimean amateur astronomer named Gennadiy Borisov discovered a moving

object against a stationary starfield and checked it against the database of the Minor Planet Center (MPC) managed by the International Astronomical Union (IAU). Borisov has previously discovered eight small bodies in the Solar System, using telescopes that he built at home. As things turned out, the particular body had not been previously observed. Borisov measured the trajectory, and found out that it had a near Earth rating of 100 percent, which meant that the object would approach the planet closely, and was a possible hazard. Borisov immediately posted an update on the database, and noted that the object had a diffuse boundary, which indicated that it was a comet and not an asteroid. As seen from the Earth, the object appears close to the Sun and that region of the sky is not usually a part of ground-based sky surveys, or NEOWISE, NASA's dedicated asteroid hunting instrument.

The object was initially designated as C/2019 Q4. Soon after the discovery by Borisov, NASA's Scout System, which is located at the Jet Propulsion Laboratory (JPL) in California, flagged the object as possibly being

interstellar in origin. NASA astronomers with their European colleagues worked together to better understand the object. A number of astronomical instruments were turned towards the object to study it. These included the William Herschel Telescope, the Gemini West Telescope, and the Gemini North Telescopes at Mauna Kea in Hawaii and even the Hubble Space Telescope. These observations helped characterise the object and determine its precise trajectory. The calculations of the trajectory showed that the comet would approach the Sun outside the orbit of Mars, and would be 300 million kilometres away at its closest approach to Earth. The closest approach will take place on Dec 28, 2019. The object is moving at a tremendous velocity – 150,000 kilometres per hour, which is the strongest indication that it has originated outside the Solar System. Davide Farnocchia from NASA's Center for Near-Earth Object Studies says, "The high velocity indicates not only that the object likely originated from outside our Solar System, but also that it will leave and head back to interstellar space". The comet is

expected to pass through the ecliptic plane – the plane roughly on which all the planets orbit the Sun – at a 40 degree angle. The object approached the Solar System from the direction of Cassiopeia, and will exit it towards the constellation of Telescopium. After millions of years of interstellar travel, the comet may make a short visit to another star system. All other known comets so far have either originated from the Kuiper belt, or the Oort cloud at a distance of one light year from the Sun.

The previous interstellar visitor, 'Oumuamua was spotted just as it was on its way out from the Solar System. There was only a narrow window of 80 days in which astronomers could observe the object. The observation campaigns that were already underway were interrupted so that the instruments could observe the fleeing body. C/2019 Q4 however, will be observable even with amateur equipment till April 2020. Professional instruments can continue to monitor the object till at least October 2020. The object is expected to peak in brightness during mid-December, 2019. Considering how close the comet appears to the Sun, it remains quite hard to observe. The object is also close to the horizon, so the observations are disturbed by ambient twilight. The conditions are expected to improve over the next few months. Because of its velocity, and the distance from the Sun, the gravity of the Sun will alter the trajectory of the comet less than 'Oumuamua.

Because of the larger observation window available on the second interstellar visitor, the object has been dubbed "the Christmas comet" by

Credit: Gemini Observatory/NSF/AURA